ARMAGEDDON

Strategy Case Competition

2026

Case Problem

Ironbridge Industrial Group

Established in 1978 and headquartered in Pune, Ironbridge Industrial Group manufactures precision auto components, industrial equipment and machinery, and specialty materials and coatings. Its customers include vehicle OEMs, capital-goods manufacturers, renewable-energy equipment makers and general engineering firms.

The three businesses serve different customer needs. Auto Components, the largest revenue contributor, supplies precision shafts, housings and brackets. Customers require dimensional accuracy, component traceability, reliable delivery and competitive pricing. Production combines forging, machining and assembly.

Industrial Equipment & Machinery supplies industrial pumps, drive modules and custom assemblies. Its customers value equipment reliability, configuration flexibility, lead times and service support. Specialty Materials & Coatings produces protective coatings and specialty compounds through mixing, batch processing and coating operations, with customers emphasising batch consistency, application performance and technical support.

An international production network

Ironbridge operates 11 plants: five in Auto Components, three in Industrial Equipment & Machinery and three in Specialty Materials & Coatings. Six plants are in India, two in Vietnam and three in Central Europe. Europe is its largest export market, followed by North America and Southeast Asia.

This footprint places different production environments within the same Group. Component lines, equipment assembly and batch processing must each meet their customers' requirements while coordinating production, materials and delivery. The new CEO must consider where a Group-wide approach is appropriate and where individual businesses require a different approach.

The digital starting point

Digital infrastructure is uneven across the network. Eight plants use a shared ERP, while three rely on local systems and monthly manual consolidation. Centrally accessible machine-level data are available for 40% of production equipment; other equipment relies on local logs or manual records. Data definitions also vary across plants.

A central team of 24 employees covers data engineering, analytics, architecture and programme delivery. Plant teams remain involved in day-to-day use of the systems: maintenance staff respond to alerts, quality teams review flagged defects and planners approve scheduling recommendations. The existing programme therefore sits across central technology capabilities and local operating practices.

Ironbridge and all company data are fictional. Financial amounts are in USD million unless stated otherwise; financial years end on 31 December.

Digital transformation and board mandate

Factory of the Future

In 2023, Ironbridge launched its debt-funded Factory of the Future programme. Predictive maintenance was intended to reduce unplanned downtime, computer-vision inspection to improve quality, and an agentic supply-chain platform to improve planning and inventory management. The initiatives were deployed across selected machines, lines and plants rather than the full production network.

The projects are at different stages of deployment. Maintenance alerts are available in production, but work-order integration remains incomplete. Vision inspection covers selected product families and still requires manual review of some flagged rejects. The planning platform recommends schedules and replenishment actions, with planners retaining approval authority.

The board now wants to understand how the programme fits the needs of the wider business. Its review must consider the existing investments alongside opportunities beyond those deployments, including how digital capabilities could support Ironbridge's customer proposition and future growth. The exhibits provide the financial, operating and project data for this assessment.

Management perspectives

Management has different views on the challenges facing Ironbridge. The COO points to input and energy costs, while the Auto Components sales head highlights OEM pricing and changing product demand. Plant managers raise uneven adoption of the new tools, and finance draws attention to programme costs. The board has not settled on a diagnosis or a preferred transformation approach.

Board intervention

Following a ratings-agency downgrade warning and shareholder pressure, the board removed the Chief Digital & Technology Officer and appointed a new CEO. The CEO has 18 months to turn the business around or face a forced sale of Auto Components. For this case, the planning period is January 2026 to June 2027, using 2025 as the baseline.

Problem statement

The new CEO has commissioned your team to develop a digital transformation strategy that strengthens Ironbridge's business performance, competitiveness and long-term growth. Your recommendations should address:

1) Identify and prioritise opportunities for digital transformation across Ironbridge's operations and wider value chain.

2) Recommend how digital capabilities can improve the existing business and create new avenues for growth and differentiation.

3) Develop a phased roadmap for the CEO's 18-month mandate, outlining the capabilities, resources and organisational changes needed for implementation.

Assess the existing digital programme and explore additional AI and automation use cases in manufacturing and the wider value chain. Recommendations are not limited to the three initiatives described; teams should determine and justify their preferred scope and priorities.

Submission guidelines and evaluation

1) Submit a PDF of up to seven slides: five main slides, including any cover slide, and two appendix slides. References must fit within this limit; external annexures and linked files will not be evaluated.

2) Present your diagnosis, prioritised recommendations, quantified impact and roadmap. Supporting calculations and references may appear in the appendix; there is no mandatory slide template.

3) Treat case data as authoritative. Where information is missing, use clearly stated assumptions supported by relevant external benchmarks from Industry data.

4) Do not include team-member names, institute names, institute logos or other identifying details.

5) Follow the deadline and timezone in the official competition platform or organiser communication. Results will be announced on 3 October 2026.

6) Generative AI tools may assist your work, but teams must independently verify all information, calculations and sources. Teams remain responsible for accuracy and originality; unverified content may adversely affect evaluation.

Evaluation criteria

Criterion	Weight
Diagnosis and analytical rigour	25%
Strategic choices and prioritisation	25%
Financial feasibility and quantified impact	25%
Implementation roadmap and risk management	15%
Clarity of thought and presentation	10%

Evaluation rewards sound reasoning, credible assumptions and consistency with the exhibits.

Exhibit 1 Income statement and operating expenses

USD million unless otherwise stated

	2022	2023	2024	2025
Revenue	1,150.0	1,205.0	1,230.0	1,190.0
Cost of Sales	860.0	910.0	945.0	930.0
Gross Profit	290.0	295.0	285.0	260.0
Operating Expenses excluding D&A	168.0	187.0	215.0	235.0
EBITDA	122.0	108.0	70.0	25.0
Depreciation & Amortisation	58.0	66.0	78.0	90.0
EBIT	64.0	42.0	(8.0)	(65.0)
Net Finance Costs	24.0	27.0	33.0	40.0
Profit Before Tax	40.0	15.0	(41.0)	(105.0)
Tax (25%)	10.0	4.0	0.0	0.0
Net Income	30.0	11.0	(41.0)	(105.0)
EBITDA Margin %	10.6%	9.0%	5.7%	2.1%
Net Margin %	2.6%	0.9%	(3.3%)	(8.8%)

Operating expense breakdown

Expense category	2022	2023	2024	2025
Non-digital personnel costs	90.0	95.0	101.0	108.0
Selling and customer support	35.0	38.0	42.0	45.0
Digital and AI operating costs	8.0	18.0	35.0	48.0
Other overheads	35.0	36.0	37.0	34.0
Total Operating Expenses	168.0	187.0	215.0	235.0

Cost of Sales and Operating Expenses exclude D&A. Digital operating costs include expensed digital personnel, software, cloud, support and implementation costs, with no overlap with other expense categories or capitalised spend. Figures are rounded, including 2023 tax of USD 4m.

Exhibit 2 Balance sheet

USD million

Assets

	2022	2023	2024	2025
Cash & Cash Equivalents	90.0	75.0	55.0	30.0
Trade Receivables	150.0	158.0	168.0	172.0
Inventory	205.0	220.0	240.0	255.0
Other Current Assets	38.0	40.0	43.0	45.0
Total Current Assets	483.0	493.0	506.0	502.0
PPE (net)	590.0	630.0	660.0	680.0
Capitalised Digital & AI Assets	65.0	100.0	135.0	155.0
Other Intangible Assets	192.0	191.0	159.0	113.0
Total Non-Current Assets	847.0	921.0	954.0	948.0
TOTAL ASSETS	1,330.0	1,414.0	1,460.0	1,450.0

Liabilities and equity

	2022	2023	2024	2025
Trade Payables	165.0	178.0	188.0	195.0
Short-term Debt	90.0	115.0	145.0	180.0
Other Current Liabilities	55.0	58.0	60.0	63.0
Total Current Liabilities	310.0	351.0	393.0	438.0
Long-term Debt	470.0	510.0	555.0	605.0
Other Long-term Liabilities	30.0	30.0	30.0	30.0
Total Non-Current Liabilities	500.0	540.0	585.0	635.0
Total Liabilities	810.0	891.0	978.0	1,073.0
Equity	520.0	523.0	482.0	377.0
TOTAL LIABILITIES & EQUITY	1,330.0	1,414.0	1,460.0	1,450.0

Digital assets are net carrying values. The USD 65m opening digital balance relates to earlier systems. Equity reflects a USD 8m dividend in 2023; no dividends were paid in 2024–2025.

Exhibit 3 Segment financials

Segment revenue and EBITDA reconcile to Group totals.

Segment revenue

	2022	2023	2024	2025
Auto Components	610.0	630.0	615.0	570.0
Industrial Equipment & Machinery	350.0	375.0	400.0	405.0
Specialty Materials & Coatings	190.0	200.0	215.0	215.0
Total Segment Revenue	1,150.0	1,205.0	1,230.0	1,190.0

Segment EBITDA

	2022	2023	2024	2025
Auto Components	68.0	58.0	30.0	5.0
Industrial Equipment & Machinery	40.0	42.0	38.0	33.0
Specialty Materials & Coatings	14.0	8.0	2.0	(13.0)
Total Segment EBITDA	122.0	108.0	70.0	25.0

Segment EBITDA margin

	2022	2023	2024	2025
Auto Components	11.1%	9.2%	4.9%	0.9%
Industrial Equipment & Machinery	11.4%	11.2%	9.5%	8.1%
Specialty Materials & Coatings	7.4%	4.0%	0.9%	(6.0%)
Group EBITDA Margin	10.6%	9.0%	5.7%	2.1%

Revenue and EBITDA are in USD million; margins are percentages. Direct project costs are charged to the relevant segments. Shared corporate and central digital costs are allocated by annual segment revenue; no corporate costs remain unallocated.

Exhibit 4 Cash flow and capital expenditure

USD million. Outflows and deductions appear in parentheses.

Cash flow item	2023	2024	2025
Net income	11.0	(41.0)	(105.0)
Add depreciation and amortisation	66.0	78.0	90.0
Increase in operating working capital	(9.0)	(21.0)	(11.0)
Operating cash flow after interest and tax	68.0	16.0	(26.0)
Maintenance PPE capex	(45.0)	(45.0)	(45.0)
Physical upgrade and expansion capex	(35.0)	(30.0)	(25.0)
Capitalised Factory of the Future spend	(50.0)	(55.0)	(45.0)
Other intangible additions	(10.0)	0.0	0.0
Total capital expenditure	(140.0)	(130.0)	(115.0)
Free cash flow before asset disposals	(72.0)	(114.0)	(141.0)
Other intangible asset disposal proceeds	0.0	19.0	31.0
Net debt raised	65.0	75.0	85.0
Dividends paid	(8.0)	0.0	0.0
Net change in cash	(15.0)	(20.0)	(25.0)
Opening cash	90.0	75.0	55.0
Closing cash	75.0	55.0	30.0

Free cash flow = operating cash flow after interest and tax less total capex, before disposals and financing. Operating working capital = receivables + inventory + other current assets − payables − other current liabilities. Interest and tax equal cash payments; there are no other cash-flow adjustments or FX movements.

Asset movement reconciliation

Depreciation or amortisation	2023	2024	2025
PPE	40.0	45.0	50.0
Digital and AI assets	15.0	20.0	25.0
Other intangible assets	11.0	13.0	15.0
Total	66.0	78.0	90.0

Other intangible disposals were non-core licences sold at carrying value; no PPE or digital assets were disposed of. Maintenance capex sustains existing operations. Justify any proposed reduction; other investment categories may be reviewed.

Exhibit 5 Digital programme economics

USD million. Project spending covers the Factory of the Future programme launched in 2023.

Project	Capitalised spend 2023–25	Net assets at end 2025	2025 operating cost	2025 gross benefit
Predictive maintenance	50.0	38.0	10.0	5.0
Computer-vision inspection	55.0	44.0	12.0	7.0
Supply-chain planning	45.0	33.0	14.0	1.0
Programme total	150.0	115.0	36.0	13.0

Project operating costs exclude D&A. The following exhibit separates recurring and temporary expenditure; shared central costs are stated separately.

Gross benefits are realised 2025 savings before operating costs, already included in Cost of Sales. They are not exit run-rates. Do not count them again as incremental future gains.

Additional investment requests

Project	Requested capex over next 18 months	Purpose
Predictive maintenance	8.0	Connect additional assets and integrate work orders
Computer-vision inspection	12.0	Add inspection stations and calibrate new part families
Supply-chain planning	10.0	Integrate systems and expand planning coverage

These capex requests are unapproved estimates, not a funding commitment. They exclude recurring operating costs and the remaining expensed implementation work shown below. Teams may prioritise existing or additional use cases.

Exhibit 6 AI project cost breakdown

Simulated 2025 expensed costs in USD million. All amounts are included in the operating costs in Exhibit 5; none is additional to the reported financials.

Cost category	Predictive maintenance	Vision inspection	Supply-chain planning
Recurring operating costs	3.0	4.0	6.0
Software and platform licences	0.8	1.0	1.5
Cloud or edge compute and data storage	0.7	1.2	1.3
Metered language-model inference / tokens	0.0	0.0	0.8
Monitoring, support and model maintenance	1.5	1.8	2.4
Temporary implementation costs	7.0	8.0	8.0
Systems integration and data engineering	4.0	4.0	5.0
Data preparation, labelling and initial model validation	2.0	3.0	2.0
Training and change management	1.0	1.0	1.0
Total 2025 project operating cost	10.0	12.0	14.0

The recurring and temporary rows are subtotals. Predictive-maintenance and vision inference compute is included in cloud/edge costs. The planning pilot uses metered language-model APIs; its token charge is separate from hosting compute and includes input/output usage and repeated agent calls. Token pricing is not applicable to every AI project.

Commitments and remaining work

Cost item	Predictive maintenance	Vision inspection	Supply-chain planning
Committed portion of annual recurring cost in first 12 months	1.0	1.5	2.0
Remaining expensed implementation work in 2026	1.0	2.0	3.0

For the case, commitments are payable during January–December 2026 even if deployment stops and are included in recurring costs above. Remaining recurring costs are avoidable only when the associated service or resource is withdrawn. From January 2027, commitments may be renewed or allowed to expire without penalty. Any shutdown or transition costs require explicit assumptions.

Temporary 2025 implementation expenditure does not automatically recur. Continuing the current scope requires the remaining expensed work above; expansion may require more. These estimates exclude the separate capex requests. Recurring costs reflect the 2025 scope, not a forecast for wider deployment. Shared central costs of USD 12m annually remain in the base case unless teams justify a feasible reduction.

Exhibit 7 Digital deployment and operational evidence

2025 results apply only to the deployed scope. Targets are original business-case targets, not achieved results.

Project and deployment	Baseline and target	2025 outcome and adoption	Reported constraint
Predictive maintenance 150 machines; two Auto Components plants and one Equipment plant	Unplanned downtime: 12% baseline; 7% target, as share of scheduled machine hours	9% downtime on the same cohort; 60% of actionable alerts closed within the required window	Incomplete maintenance records and work-order integration; technician availability
Vision inspection Eight lines; two Auto Components plants and one Materials plant	Scrap/rework: 6.0% baseline; 3.5% target, as share of covered-line production cost	4.8% scrap/rework; 45% of flagged rejects required manual review	Calibration when product specifications change; selected part families only
Supply-chain planning Two Auto Components plants and one Equipment plant; planners approve actions	Pilot inventory days: 90 baseline; 75 target. On-time delivery: 86% baseline	94 inventory days; 86% on-time delivery; 35% of schedules accepted without override	Master-data inconsistencies and separate planning systems

Inventory days = average pilot inventory / annual pilot cost of sales × 365. Planning benefits of USD 1m reflect lower expediting costs, not inventory release. Local observations do not establish causality or guarantee results elsewhere. Use explicit, benchmark-supported assumptions for extrapolation and new use cases; separate inventory cash release from recurring EBITDA gains.

Exhibit 8 Operating context

Simulated 2025 company data. Financial amounts are in USD million.

Manufacturing footprint and operating costs

Measure	Auto Components	Equipment & Machinery	Materials & Coatings
Plants by location	5: India 3; Vietnam 1; Central Europe 1	3: India 1; Vietnam 1; Central Europe 1	3: India 2; Central Europe 1
Main processes	Machining, forging and assembly	Fabrication, machining and assembly	Mixing, batch processing and coating
Capacity utilisation	72%	81%	67%
Energy expenditure	45.0	25.0	50.0
Direct production labour	95.0	60.0	25.0

Capacity utilisation is actual output as a share of practical capacity at the 2025 product mix. Energy and direct production labour are separate components of the USD 930m Cost of Sales; neither is included in Operating Expenses.

Digital readiness

Area	Current position
ERP coverage	8 of 11 plants use a shared ERP; 3 use local systems with monthly manual consolidation.
Machine data	40% of production equipment has centrally accessible machine-level data. Other equipment relies on local logs or manual records; data definitions vary across plants.
Central digital team	24 employees covering data engineering, analytics, architecture and programme delivery. Their payroll is included in the USD 12m shared central digital cost.

Commercial context

Segment	Representative products	Principal customer requirements
Auto Components	Precision shafts, housings and brackets	Dimensional accuracy, traceability, reliable delivery and competitive pricing
Equipment & Machinery	Industrial pumps, drive modules and custom assemblies	Reliability, configuration flexibility, lead times and service support
Materials & Coatings	Protective coatings and specialty compounds	Consistent batches, application performance and technical support